

B.TECH COMPUTER SCIENCE & ENGINEERING

(Artificial Intelligence & Machine Learning)

First Year

Semester I

Subject Code	Subject Name	Credit
BAS101 / BAS102	Engineering Physics / Engineering Chemistry	4
BAS103	Engineering Mathematics-I	4
BEE101 / BEC101	Fundamentals of Electrical Engineering / Fundamentals of Electronics Engineering	3
BCS101 / BME101	Programming for Problem Solving / Fundamentals of Mechanical Engineering	3
BAS104 / BAS105	Environment and Ecology / Soft Skills	3
BAS151 / BAS152	Engineering Physics Lab / Engineering Chemistry Lab	1
BEE151 / BEC151	Basic Electrical Engineering Lab / Basic Electronics Engineering Lab	1
BCS151 / BAS155	Programming for Problem Solving Lab / English Language Lab	1
BCE151 / BWS151	Engineering Graphics & Design Lab / Workshop Practice Lab	2
Total Credits		22

Semester II

Subject Code	Subject Name	Credit
BAS202 / BAS201	Engineering Chemistry / Engineering Physics	4
BAS203	Engineering Mathematics-II	4
BEC201 / BEE201	Fundamentals of Electronics Engineering / Fundamentals of Electrical Engineering	3
BME201 / BCS201	Fundamentals of Mechanical Engineering / Programming for Problem Solving	3
BAS205 / BAS204	Soft Skills / Environment and Ecology	3
BAS252 / BAS251	Engineering Chemistry Lab / Engineering Physics Lab	1
BEC251 / BEE251	Basic Electronics Engineering Lab / Basic Electrical Engineering Lab	1
BAS255 / BCS251	English Language Lab / Programming for Problem Solving Lab	1
BWS251 / BCE251	Workshop Practice Lab / Engineering Graphics & Design Lab	2
BVA251 / BVA252	Sports and Yoga / NSS (Qualifying Course)	0
Total Credits		22

Second Year

Semester III

Subject Code	Subject Name	Credit
BOE3** / BAS303	Science Based Open Elective / Mathematics-III / IV / V / Universal Human Value	4
BVE301 / BAS301	Universal Human Value & Professional Ethics / Technical Communication	3
BCS301	Data Structure	4
BCS302	Computer Organization and Architecture	4
BCS303	Discrete Structures & Theory of Logic	3
BCS351	Data Structure Lab	1
BCS352	Computer Organization and Architecture Lab	1
BCS353	Web Designing Workshop	1
BCC301 / BCC302	Cyber Security / Python Programming	2
BCC351	Internship Assessment / Mini Project	2
Total Credits		25

Semester IV

Subject Code	Subject Name	Credit
BAS403 / BOE4**	Mathematics-III / IV / V / Science Based Open Elective	4
BAS401 / BVE401	Technical Communication / Universal Human Value & Professional Ethics	3
BCS401	Operating System	4
BCS402	Theory of Automata and Formal Languages	4
BCS403	Object Oriented Programming with Java	3
BCS451	Operating System Lab	1
BCS452	Object Oriented Programming with Java Lab	1
BCS453	Cyber Security Workshop	1
BCC402 / BCC401	Python Programming / Cyber Security	2
BVE451 / BVE452	Sports and Yoga-II / NSS-II	0
Total Credits		23

Third Year

Semester V

Subject Code	Subject Name	Credit
BCS501	Database Management System	4
BCAI501	Artificial Intelligence	4
BCS503	Design and Analysis of Algorithm	4
BCAI051 / BCS058 / BCS052 / BCS054	Departmental Elective-I	4
BCAM051 / BCAI052 / BCS056 / BCS057	Departmental Elective-II	4
BCS551	Database Management System Lab	1
BCAI551	Artificial Intelligence Lab	1
BCS553	Design and Analysis of Algorithm Lab	1
BCS554	Mini Project / Internship Assessment	2
BNC501 / BNC502	Constitution of India / Essence of Indian Traditional Knowledge	0
Total Credits		23

Semester VI

Subject Code	Subject Name	Credit
BCS601	Software Engineering	4
BCAI601	Machine Learning Techniques	4
BCS603	Computer Networks	4
BCAI061 / BCDS061 / BCAM061 / BCAM062	Departmental Elective-III	3
Open Elective-I	Open Elective-I	3
BCS651	Software Engineering Lab	1
BCAI651	Machine Learning Lab	1
BCS653	Computer Networks Lab	1
BNC601 / BNC602	Constitution of India / Essence of Indian Traditional Knowledge	0
Total Credits		21

Fourth Year

Semester VII

Subject Code	Subject Name	Credit
BAI701	Deep Learning	3
B**07*	Departmental Elective-IV	3
BOEM**	Open Elective-II	3
BAI751	Deep Learning Lab	1
BCS752	Mini Project / Internship Assessment	2
BCS753	Project-I	5
BCS754	Startup & Entrepreneurial Activity Assessment	2
Total Credits		19

Semester VIII

Subject Code	Subject Name	Credit
BOENM**	Open Elective-III	3
BOENM**	Open Elective-IV	3
BCS851	Project-II	10
Total Credits		16

Departmental Electives

Departmental Elective-I

- **BCAI051** – Mathematical Foundation of AI, ML and Data Science
- **BCS058** – Data Warehouse & Data Mining
- **BCS052** – Data Analytics
- **BCS054** – Object Oriented System Design with C++

Departmental Elective-II

- **BCAM051** – Cloud Computing
- **BCAI052** – Natural Language Processing
- **BCS056** – Application of Soft Computing
- **BCS057** – Image Processing

Departmental Elective-III

- **BCAI061** – Cyber Forensic Analytics
- **BCDS061** – Image Analytics
- **BCAM061** – Social Media Analytics and Data Analysis
- **BCAM062** – Stream Processing and Analytics

Departmental Elective-IV

- **BCS070** – Internet of Things
- **BCS072** – Cryptography & Network Security
- **BAI070** – Principles of Generative AI
- **BAI071** – Blockchain Architecture Design